

Preliminary Economic Modeling of an External Anatomical Sacral Barrier for Contamination-Driven Dressing Failure

A cost-consequence and budget impact analysis for sacral pressure injuries, fecal incontinence, and recurrent dressing loss caused by fecal contamination.

Model type

Preliminary decision-analytic model with hospital institutional perspective.

Primary operational endpoint

Contamination-attributable dressing changes per patient-month.

Editorial structure

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Background, objective, methods, findings and interpretation.

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05. Model Structure

Decision logic, formulas and break-even threshold.

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Why fecal contamination is an operational failure mode.

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Perspective, population, comparators and time horizon.

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08. Discussion

Clinical meaning, nursing workload and research priorities.

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Traceable scientific and health-economic sources.

This article is formatted for website publication and institutional communication. It preserves a conservative scientific position: modeled assumptions are not presented as observed clinical efficacy. Real-world effectiveness is planned for measurement during the commercial post-market phase.

Preliminary Cost-Consequence and Budget Impact Modeling of an External Anatomical Sacral Barrier for Contamination-Driven Dressing Failure in Sacral Pressure Injuries

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Abstract

Background: Sacral pressure injuries in patients with fecal incontinence are frequently complicated by contamination of the dressing interface. Liquid or semi-formed stool may track cranially along the intergluteal cleft toward the coccyx and dressing border. Absorbent products are essential components of care, but they do not necessarily interrupt this anatomical pathway mechanically.

Objective: To estimate the economic plausibility of adding an external anatomical sacral barrier to standard care for adults with sacral pressure injuries and fecal incontinence, using a preliminary cost-consequence and budget impact framework.

Methods: A decision-tree model was structured from the hospital institutional perspective. The simulated population included adult patients with stage III-IV sacral pressure injuries, fecal incontinence, and recurrent contamination-driven dressing loss. Two strategies were compared over a 30-day horizon: standard care alone and standard care plus an external anatomical sacral barrier. Costs included advanced dressings, nursing labor, personal protective equipment, consumables and device acquisition. The unit price considered for SWB in the initial commercial phase was R\$5.00.

Results: The principal economic driver was the number of contamination-attributable dressing changes avoided per patient-month. With a device cost of R\$5.00 and estimated use of 2-4 units/day, monthly device cost ranges from approximately R\$300 to R\$600 per patient. If a dressing episode costs R\$350, the device would need to avoid approximately 0.9-1.7 dressing changes per month to reach cost neutrality.

Conclusion: This preliminary model suggests that an external anatomical sacral barrier may be economically plausible in patients with recurrent fecal contamination of sacral dressings. Clinical effectiveness is not assumed as established; real-world effectiveness will be measured during post-market use through structured institutional follow-up.

Keywords: Sacral pressure injury; fecal incontinence; dressing failure; contamination; anatomical barrier; budget impact; cost-consequence analysis; nursing workload.

Introduction

The sacral region combines pressure, shear, friction, local heat, moisture and occlusion. In bedridden patients with fecal incontinence, the wound environment can be repeatedly destabilized by exposure to fecal effluent. When contamination reaches the dressing border or the periwound region, it may compromise adhesion, reduce dressing dwell time, increase maceration and force unplanned replacement.³

A clinically relevant mechanism is fecal tracking along the natal or intergluteal cleft. Absorbent products reduce exposure volume and remain foundational for hygiene, but they do not necessarily obliterate the anatomical space through which loose or semi-formed stool can migrate toward the coccyx and sacral dressing edge.

This mechanism creates a recurring operational cycle: contamination, cleansing, reassessment, re-dressing and renewed risk of contamination. For hospitals and long-term care institutions, this cycle consumes advanced dressings, personal protective equipment, cleansing supplies and nursing time. For patients, it may increase manipulation, discomfort and instability of fragile periwound skin.^{3,8}

Fecal contamination should be treated as a measurable dressing-failure mode, not merely as a hygiene event.

Objective

The objective was to estimate whether adding an external anatomical sacral barrier to standard care is economically plausible in adult patients with sacral pressure injuries and fecal incontinence. The model identifies the conditions under which the device may become cost-neutral or cost-saving by reducing contamination-attributable dressing replacement.

Clinical and technical rationale

The SWB is described as a disposable, non-invasive external barrier designed to fit the intergluteal cleft and protect sacral wounds, dressings and medical devices from fecal and urinary contamination. Its technical description identifies it as a Class I medical device, notified to ANVISA under number 80220130014, made primarily of medical hydrophobic polyurethane foam.^{9,10}

Methods

Study design

A preliminary decision-analytic model was developed using a cost-consequence and budget impact structure. The model was guided by health-economic reporting principles and decision-model transparency standards.^{1,2}

Perspective and horizon

The analysis used the hospital institutional perspective. The base-case horizon was 30 days, which is operationally appropriate for measuring dressing consumption, device use, nursing workload and contamination-attributable re-dressing events.

Population

The modeled population included adult patients with stage III-IV sacral pressure injuries or complex sacral wounds, fecal incontinence or frequent loose/semi-formed stool, recurrent dressing contamination and use of diapers or absorbent products as part of routine care.

Comparators

Standard care

Advanced dressing according to institutional protocol, cleansing, topical barriers when indicated, diapers or absorbent products, repositioning, pressure redistribution and nursing assessment.^{3,5,6}

Standard care + SWB

All elements of standard care plus an external anatomical barrier positioned in the intergluteal cleft. The device is modeled as complementary, not as a replacement for pressure injury prevention or advanced dressings.

Primary endpoint

The primary operational endpoint was the number of contamination-attributable dressing changes per patient-month. The primary economic endpoint was total direct monthly cost per patient.

Model Structure

The model estimates total monthly cost per patient from dressing episodes, device use and optional scenario-based complication costs. The principal economic mechanism is avoided premature dressing replacement.

Total monthly cost = dressing episode costs + monthly device cost + scenario-specific complication costs

Dressing episode costs = number of dressing changes × cost per dressing episode

Monthly device cost = device unit price × devices used per day × 30 days

Incremental cost = total cost with SWB - total cost with standard care

A negative incremental cost indicates potential savings. A positive incremental cost indicates added cost. Cost neutrality occurs when the value of avoided dressing episodes equals the monthly cost of the device.

Break-even logic

Required avoided dressing changes = monthly device cost ÷ cost per dressing episode

With SWB priced at R\$5.00 per unit and estimated use of 2-4 units/day, monthly device cost ranges from approximately R\$300 to R\$600 per patient. If each dressing episode costs R\$350, the device must avoid approximately 0.9-1.7 dressing changes per month to reach cost neutrality. Additional avoided dressing changes represent potential net savings.

R\$5
Unit price

Initial commercial phase.

2-4
Units/day

Modeled use range.

**0.9-
1.7**
Changes/month

Break-even estimate at R\$350
per dressing episode.

Model Parameters

The parameters below provide the structure for the economic model. Institutional implementation should replace these values with local purchasing data, nursing time-motion estimates and dressing-audit results.

Parameter	Value or range	Role in the model
Monthly dressing changes, standard care	10-60 per patient-month	Defines baseline burden of dressing loss.
Reduction in contamination-attributable dressing loss	0-70%	Sensitivity variable; not observed clinical efficacy.
Dressing cost	R\$150-350/change	Direct material cost component.
Labor, PPE and consumables	R\$200-500/change	Operational cost per dressing episode.
SWB unit price	R\$5.00	Device acquisition cost in the initial commercial phase.
SWB use	2-4 units/day	Defines monthly device cost: approximately R\$300-600/patient-month.
Additional hospitalization	Scenario only	Not assumed to be avoided in the base case.

Sensitivity Analysis

The model should be interpreted through scenario analysis. Conservative, base-case, favorable and worst-case scenarios are recommended. The most influential variables are baseline contamination frequency, cost per dressing episode, device use per day, device price and the real reduction in contamination-attributable dressing changes.

Post-market evidence plan: During commercialization, real-world effectiveness will be measured through institutional follow-up, including dressing contamination events, avoided dressing changes, dressing dwell time, daily SWB consumption, usability, device displacement, skin integrity and nursing-team perception.

Results

The model indicates that the potential economic value of SWB depends primarily on whether it reduces contamination-attributable dressing replacement in patients with recurrent fecal tracking toward the sacral dressing border. The lower the device price and the higher the cost per dressing episode, the fewer avoided dressing changes are required for cost neutrality.

Operational endpoint

The most important endpoint is contamination-attributable dressing changes per patient-month. This endpoint is directly observable, clinically meaningful and economically linked to material consumption and nursing workload.

Budget impact

Institutional budget impact can be estimated by multiplying incremental monthly cost or savings per patient by the number of eligible patients.

Monthly institutional budget impact = incremental monthly cost per patient × number of eligible patients

For a cohort of 30 high-risk patients, even a small reduction in unplanned dressing episodes can produce meaningful aggregate savings if dressing episode costs are high and contamination events are frequent.

Interpretation

The model should not be read as proof of clinical effectiveness. It is a structured economic rationale showing where real-world evidence should be collected and which endpoints matter most.

The correct economic question is not only “How much does the device cost?” but “How many contaminated dressing changes must be avoided for the device to pay for itself?”

Discussion

This model reframes fecal contamination of sacral dressings as an operationally measurable failure mode. When fecal effluent reaches the dressing border and forces premature replacement, the institution incurs direct costs with dressings, PPE, cleansing supplies, disposal and nursing time.

The economic rationale for an external anatomical barrier is based on interrupting a specific pathway: fecal tracking through the intergluteal cleft. This does not replace advanced dressings, pressure redistribution, repositioning, skin care or fecal incontinence management. It aims to preserve the therapeutic environment already established by standard protocols.^{5,6}

The model also highlights a documentation gap. Many institutions record dressing replacement without specifying whether it was planned, clinically indicated, or triggered by fecal contamination. Without this distinction, the cost of the problem remains partially invisible.

Clinical Practice Implications

Institutions evaluating SWB should document fecal contamination as a specific reason for dressing failure, measure actual dressing dwell time in incontinent patients, distinguish planned from unplanned dressing changes, and quantify nursing time associated with re-dressing events.

Research Priorities

During commercialization, post-market follow-up should measure real-world effectiveness in institutional settings. Data collection should include contamination frequency, dressing dwell time, avoided dressing changes, daily device consumption, usability, displacement, skin integrity and nursing-team perception.^{1,2,3}

Limitations

This analysis is a preliminary model. The clinical effectiveness of the external anatomical barrier will be measured through post-market follow-up during commercialization. Until these data are available, reductions in contamination-attributable dressing loss remain modeled assumptions, not observed clinical efficacy.

Dressing costs, PPE costs, labor costs, device consumption and hospitalization costs vary across institutions. Local application of the model requires replacement of assumptions with actual procurement, microcosting and clinical-audit data.

The 30-day horizon is appropriate for operational cost modeling but is not sufficient to infer wound healing, infection prevention, hospital length-of-stay reduction or quality-of-life improvement.

There is intellectual and/or commercial interest related to the device concept. Transparency, structured data collection and independent review are important safeguards when interpreting model outputs.

Conclusion

Fecal contamination of sacral pressure injury dressings is a clinically plausible and operationally measurable driver of premature dressing replacement. In patients with recurrent contamination, an external anatomical sacral barrier priced at R\$5.00 per unit may be economically plausible, particularly when advanced dressings and nursing time are costly.

The central endpoint is simple: contamination-attributable dressing changes per patient-month. If the device avoids enough unplanned dressing episodes to offset its monthly cost, it may become cost-neutral or cost-saving. Real-world post-market measurement is required to determine actual effectiveness, safety, usability and institutional impact.

References

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Editorial note: Device-specific clinical effectiveness is intentionally not presented as established. The document supports a transparent post-market measurement strategy for real-world effectiveness.